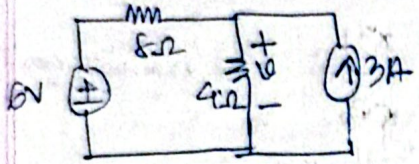
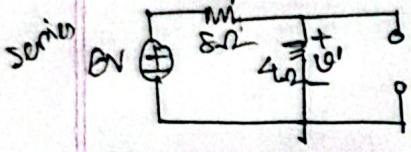
Superposition

example 4.3



Find v using superposition theorem.

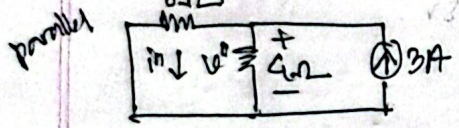
When only 6V is active,



$v' = \frac{4}{4+8} \times 6$

$= 2v$

When only 3A is active,



$i'' = \frac{8}{8+4} \times 3$

$= 2A$

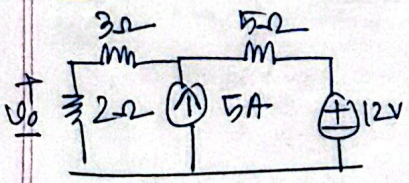
$v'' = i'' \times 4 = 2 \times 4 = 8v$

∴ total voltage, $v = v' + v''$

$= 2 + 8$

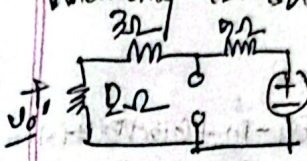
$= 10v.$

practice problem 4.3



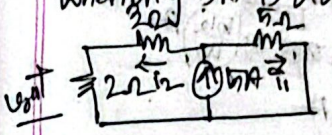
Find v_o using superposition theorem.

When only 12V is active,



$V_o' = \frac{2}{2+3+5} \times 12 = 2.4V.$

When only 5A is active,



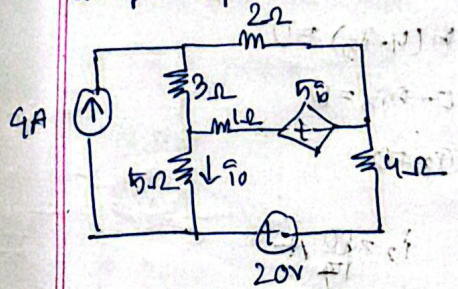
$i_2 = \frac{5}{5+3+2} \times 5 = 2.5A.$

$V_o'' = 2 \times 2.5 = 5V.$

total voltage, $v = v_o' + v_o''$

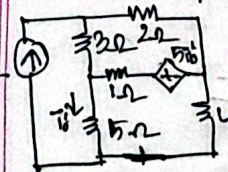
$= 2.4 + 5 = 7.4V.$

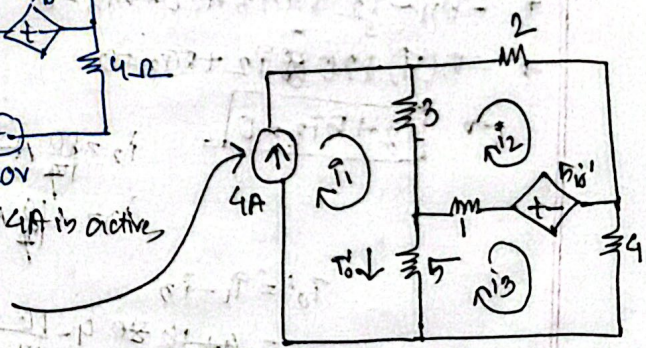
example 4.4



Find i_o

When only 4A is active,





using mesh analysis,

For mesh 1,

$i_1 = 4A.$

$i_o' = i_1 - i_3 = 4 - i_3$

For mesh 2,

$3(i_2 - i_1) + 2i_2 - 5i_o' + 1(i_2 - i_3) = 0$

$\Rightarrow -3i_1 + 6i_2 - i_3 - \frac{4}{5}(4 - i_3) = 0$

$\Rightarrow -3i_1 + 6i_2 + 4i_3 = 20 + 3i_1$

$= 20 + 3 \times 4$

$= 32$

For mesh 3,

$5(i_3 - i_2) + 1(i_3 - i_2) + 5i_0' = 0$

$\Rightarrow -6i_2 + 6i_3 + 5(4 - i_2) = 0$ $\quad -5i_1 - i_2 + 6i_3 + 5(4 + i_3) = 0$

$\Rightarrow -6i_2 + i_3 = -20$ $\quad \Rightarrow -5 \times 4 - i_2 + 6i_3 + 20 - 5i_3 = 0$

$\Rightarrow -i_2 + 6i_3 = 0 \rightarrow$ [5]

So,

i_2 is const.		
6	4	32
-1	1	20

$i_2 = 3.789\text{A} \approx \left(\frac{536}{141}\right)\text{A}$

$i_3 = 2.4\text{A}$

$i_0' = i_1 - i_3 = 4 - \frac{536}{141} = \frac{4}{141}$

$5(i_3 - i_1) + 1(i_3 - i_2) + 5i_0' + 4i_3 = 0$

$\Rightarrow 6i_3 - 5i_1 - i_2 + 4i_3 + 5(4 - i_3) = 0$

$\Rightarrow -5i_1 - i_2 + 10i_3 + 20 - 5i_3 = 0$

$\Rightarrow -5(4) + 20 - i_2 + 5i_3 = 0$

$\Rightarrow \boxed{i_2 + 5i_3 = 0}$

$i_2 = \frac{80}{17}\text{A}$

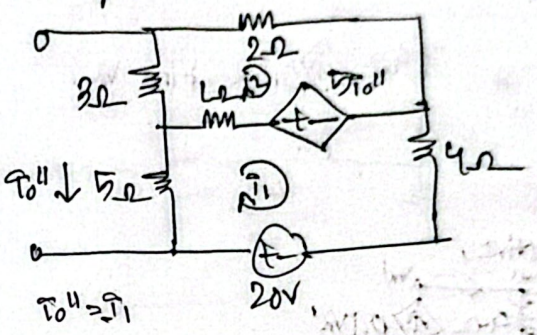
$i_3 = +\frac{16}{17}\text{A}$

$i_0' = i_2 - i_3$

$= \frac{4 + 16}{17} - \frac{80}{17} \quad 4 - \frac{16}{17}$

$= \frac{52}{17}\text{A}.$

When only 20V is active,



$i_0'' = i_1$

For mesh 1,

$i_0'' = i_1$

$-20 + 5i_1 + 1(i_1 - i_2) + 5i_0'' + 4i_1 = 0$

$\Rightarrow -20 + 5i_1 + i_1 - i_2 + 5i_1 + 4i_1 = 0$

$\Rightarrow 15i_1 - i_2 = 20$

$-5i_1 + i_2 = -20$

For mesh 2,

$3i_2 + 2i_2 - 5i_0'' + 1(i_2 - i_1) = 0$

$\Rightarrow 3i_2 + 5i_2 + 5i_1 + i_2 - i_1 = 0$

$\Rightarrow -6i_1 + 6i_2 = 0 \quad 4i_1 + 6i_2 = 0$

$\therefore -i_1 + i_2 = 0$

Solving we get,

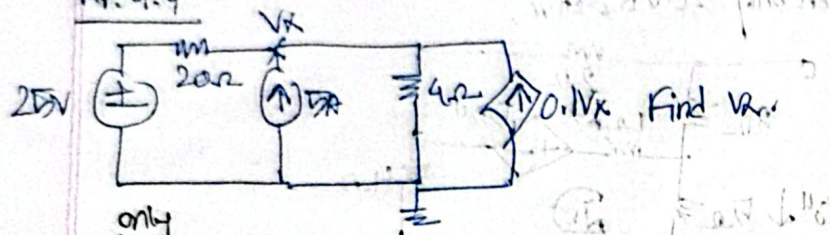
$i_1 = \frac{60}{17}\text{A} \quad \therefore i_0'' = -\frac{60}{17}\text{A}$

$i_2 = -\frac{40}{17}\text{A}$

$\therefore i_{total} = i_0'' + i_0'' = \frac{52}{17} - \frac{60}{17}$

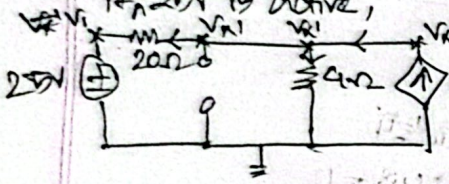
$= \frac{-8}{17} = -0.47\text{A}$

P.P. 4.4



Find V_x.

only 25V is active,



$V_1 = 25V$.

For V_x',

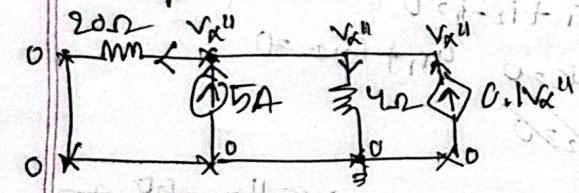
$$\frac{V_x' - V_1}{20} + \frac{V_x' - 0}{4} = 0.1V_x'$$

$$\Rightarrow V_x' - V_1 + 5V_x' = 20 \times 0.1V_x'$$

$$\Rightarrow 6V_x' - 2V_x' = 25$$

$$\therefore V_x' = \frac{25}{4} = 6.25V.$$

only 5A is active,



$$\frac{V_x'' - 0}{20} + \frac{V_x'' - 0}{4} = 5 + 0.1V_x''$$

$$\Rightarrow V_x'' + 5V_x'' = 100 + 2V_x''$$

$$\Rightarrow 4V_x'' = 100$$

$$\therefore V_x'' = 25V$$

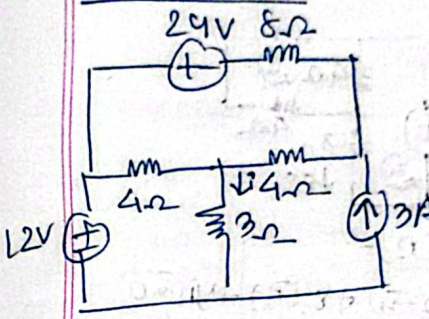
∴ total voltage,

$V_x = V_x' + V_x''$

$= 25 + 6.25$

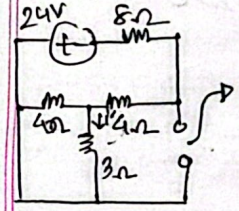
$= 31.25V.$

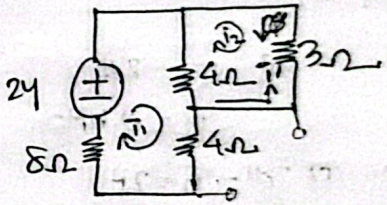
example 4.5



Find i using superposition.

If 24V is active only,





For mesh 1,

$$8i_1 - 24 + 4(i_1 - i_2) + 4i_1 = 0$$

$$\Rightarrow 16i_1 - 4i_2 = 24$$

$$\Rightarrow \boxed{4i_1 - i_2 = 6}$$

For mesh 2,

$$4(i_2 - i_1) + 3i_2 = 0$$

$$\Rightarrow 4i_2 - 4i_1 + 3i_2 = 0$$

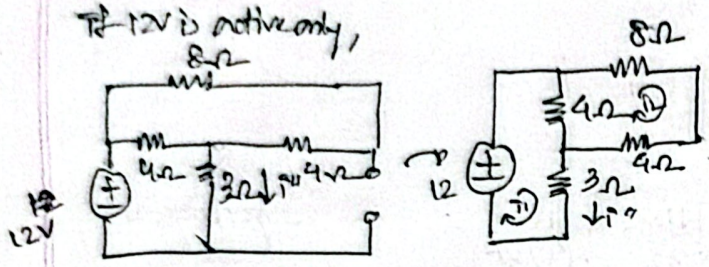
$$\Rightarrow \boxed{-4i_1 + 7i_2 = 0}$$

$$i_1 = 1.75A$$

$$i_2 = 1A$$

$$\therefore i = -i_2 = -1A$$

If 12V is active only,



mesh 1,

$-12+(4+3)i_1-4i_2=0$

$\Rightarrow 7i_1-4i_2=12$

mesh 2,

$4(i_2-i_1)+8(i_2)+4i_2=0$

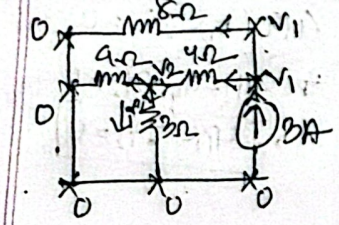
$\Rightarrow -4i_1+16i_2=0$

So, $i_1=2A$

$i_2=0.5A$

now, $i'=i_1=2A$

If 3A is active only,



For V_1,

$\frac{V_1-V_2}{4}+\frac{V_1-0}{8}=3$

$\Rightarrow 2V_1-2V_2+V_1=24$

$\Rightarrow 3V_1-2V_2=24$ 24

For V_2,

$\frac{V_2-0}{4}+\frac{V_2-0}{3}+\frac{V_2-V_1}{4}=0$

$\Rightarrow 3V_2+4V_2+3V_2-3V_1=0$

$\Rightarrow -3V_1+10V_2=0$

So, $V_1=10V$

$V_2=3V$

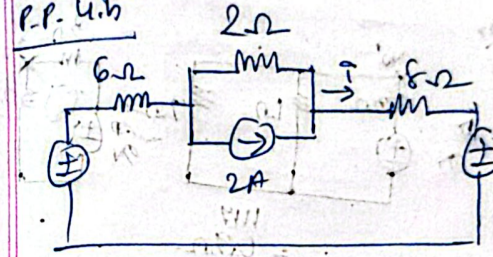
$i''=\frac{V_2-0}{3}=\frac{3-0}{3}=1A$

total, $i=i'+i''+i'''$

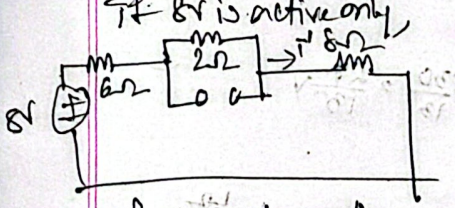
$=1+2+1$

$=2A$.

P.P. 4.5

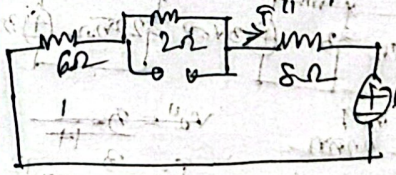


If 8V is active only,



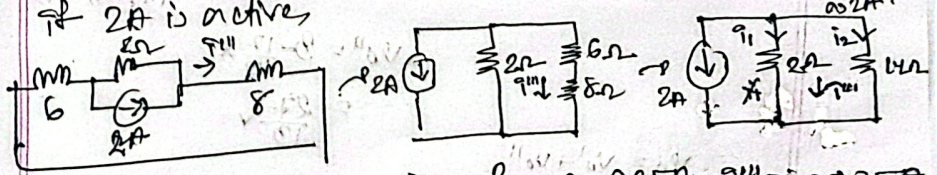
$i'=\frac{8}{6+2+8}=0.5A$

If 6V is active,



$i''=-\frac{6}{6+2+8}=-\frac{3}{8}A$.

If 2A is active,

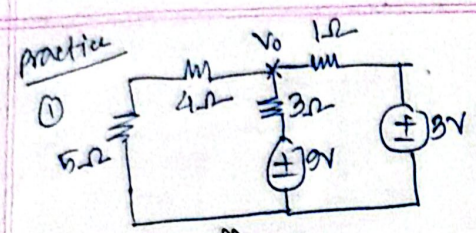


$i_2=\frac{2}{2+4}\times 2=0.25A$, $i'''=i_2=0.25A$

$I=i'+i''+i'''=0.5-\frac{3}{8}+0.25=0.375A$

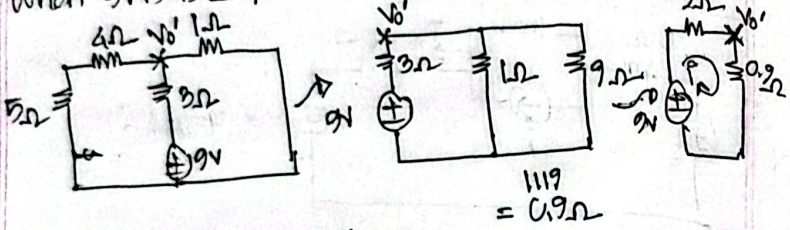
$=375mA$.

Practice

① 

Find V_o using superposition theorem

When 9V is on,



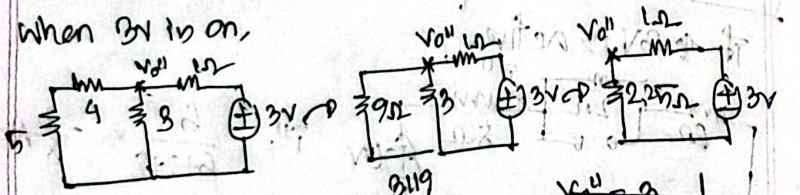
$1||9 = 0.9\Omega$

$$i = \frac{9}{3+0.9} = \frac{30}{13}\,A$$

$$V_o' = 9 - 3i = 9 - 3\times\frac{30}{13} = \frac{27}{13}\,V$$

(9 — supply, 3 — drop)

When 3V is on,



$3||9 = 2.25\Omega$

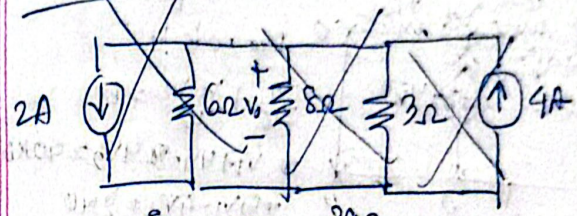
$$V_o'' = 3 - \frac{3}{1+2.25} = \frac{12}{13}\,A$$

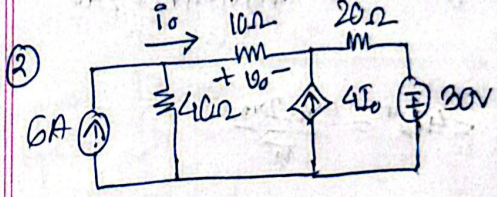
$$V_o'' = 3 - \frac{12}{13}\times 1 = \frac{27}{13}\,V$$

$$\therefore V_o = V_o' + V_o''$$

$$= \frac{27}{13} + \frac{27}{13} = \frac{54}{13} = 4.154\,V.$$

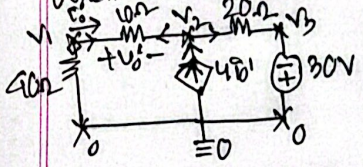
example 4.6



② 

Find i_o and V_o using superposition.

When 30V is active,



Let's use nodal analysis.

$V_3 - 0 = -30$

$\therefore V_3 = -30$

$$i_o' = \frac{V_1 - V_2}{10}$$

$$4i_o' = \frac{2(V_1 - V_2)}{5} = \frac{2V_1 - 2V_2}{5}$$

Now, at node 1,

$$\frac{V_1 - 0}{40} + \frac{V_1 - V_2}{10} = 0$$

$$\Rightarrow V_1 + 4V_1 - 4V_2 = 0$$

$$\Rightarrow 5V_1 - 4V_2 = 0$$

For node 2,

$$\frac{V_2 - V_1}{10} + \frac{V_2 - V_3}{20} = 4i_o'$$

$$\Rightarrow \frac{2V_2 - 2V_1 + V_2 - V_3}{20} = \frac{2V_1 - 2V_2}{5}$$

$$\Rightarrow -2V_1 + 3V_2 - (-30) = 8V_1 - 8V_2$$

$$\Rightarrow -10V_1 + 11V_2 = -30$$

$$\times 8$$

So, $V_1 = \frac{48}{19}\,V$, $V_2 = \frac{-60}{19}\,V = -10\,V$

$$i_o' = \frac{V_1 - V_2}{10} = \frac{\frac{48}{19} - \left(\frac{-60}{19}\right)}{10} = \frac{\frac{48+60}{19}}{10} = \frac{108}{190} = \frac{2}{10} = 0.2$$

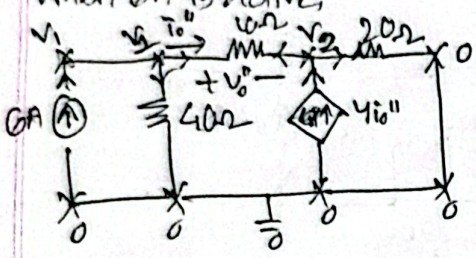
$$V_o' = V_1 - V_2 = \frac{48}{19} - \frac{-60}{19}\,V = -8 + 10 = 2V.$$

$v_o'' = v_1 - v_2$

$i_o'' = \frac{v_1 - v_2}{10}$

$4i_o'' = \frac{2v_1 - 2v_2}{5}$

When 6A is active,



For node 1,

$\frac{v_1}{40} + \frac{v_1 - v_2}{10} = 6$

$\Rightarrow v_1 + 4v_1 - 4v_2 = 40 \times 6$

$\Rightarrow 5v_1 - 4v_2 = 240$

For node 2,

$\frac{v_2 - v_1}{50} + \frac{v_2}{20} = 4i_o'' = \frac{2v_1 - 2v_2}{5}$

$\Rightarrow \frac{2v_2 - 2v_1 + v_2}{20} = \frac{2v_1 - 2v_2}{5}$

$\Rightarrow -2v_1 + 3v_2 = 8v_1 - 8v_2$

$\Rightarrow 10v_1 - 11v_2 = 0$

So, $v_1 = 176\,V$ $\quad v_2 = 160\,V$

$v_o'' = 176 - 160 = 16\,V$

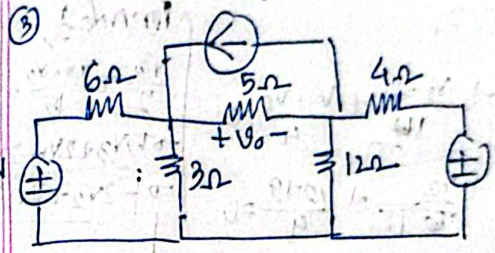
$i_o'' = \frac{16}{10} = 1.6\,A$

So, $i_o = i_o' + i_o'' = \frac{0.2}{5} + 1.6 = 1.64\,A \approx 1.6\,A$

$v_o = v_o' + v_o'' = \frac{2}{15} + 16 = 15.37\,V \approx 16\,V$

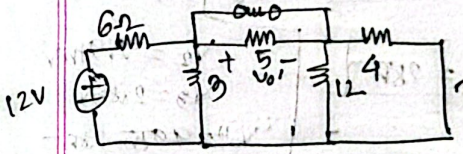
2A

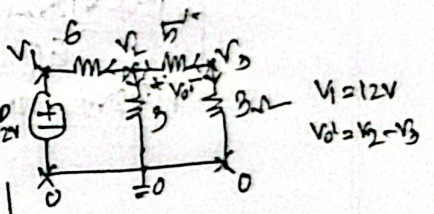
(3)



Find v_o using superposition.

When only 12V is active,





$v_1 = 12V$

$v_o' = v_2 - v_3$

12 || 4

$= \frac{4 \times 12}{12+4}$

$= \frac{48}{16} = 3\Omega$

For node 2,

$\frac{v_2 - v_1}{6} + \frac{v_2}{3} + \frac{v_2 - v_3}{5} = 0$

$\Rightarrow \frac{5v_2 - 5v_1 + 10v_2 + 6v_2 - 6v_3}{30} = 0$

$\Rightarrow -5v_1 + 21v_2 - 6v_3 = 0$

$\Rightarrow 21v_2 - 6v_3 = 5v_1 = 5 \times 12 = 60$

$\Rightarrow \boxed{7v_2 - 2v_3 = 20}$

For node 3,

$\frac{v_3 - v_2}{5} + \frac{v_3}{3} = 0$

$\Rightarrow \frac{3v_3 - 3v_2 + 5v_3}{15} = 0$

$\Rightarrow \boxed{-3v_2 + 8v_3 = 0}$

$v_2 = 3.2V$

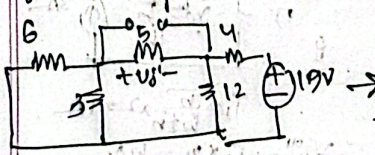
$v_3 = 1.2V$

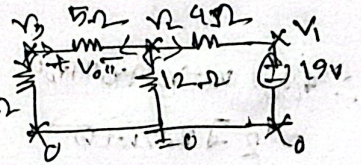
$v_o' = v_2 - v_3$

$= 3.2 - 1.2$

$= 2V$

When only 19V is active,





$6 \| 3 = \frac{3 \times 6}{3+6} = \frac{18}{9} = 2\Omega$

$v_o'' = v_3 - v_2$

$V_1 = 12V$.

$V_2 > V_3$

For node 2,

$$\frac{V_2 - V_3}{5} + \frac{V_2 - 0}{12} + \frac{V_2 - V_1}{4} = 0$$

$$\Rightarrow \frac{V_2 - V_3}{5} + \frac{V_2}{12} + \frac{V_2 - 12}{4} = 0$$

$$\Rightarrow \frac{12V_2 - 12V_3 + 5V_2 + 15V_2 - 285}{60} = 0$$

$$\Rightarrow \boxed{32V_2 - 12V_3 = 285}$$

For node 3,

$$\frac{V_3}{2} + \frac{V_3 - V_2}{5} = 0$$

$$\Rightarrow 5V_3 + 2V_3 - 2V_2 = 0$$

$$\Rightarrow \boxed{-2V_2 + 7V_3 = 0}$$

So,

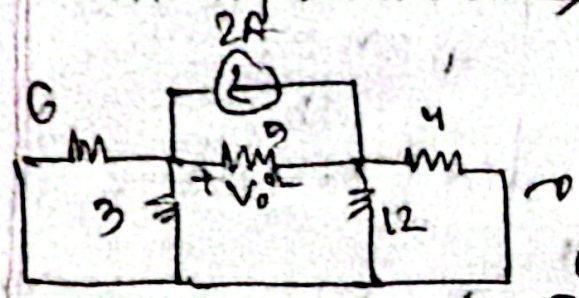
$V_2 = 9.975V$

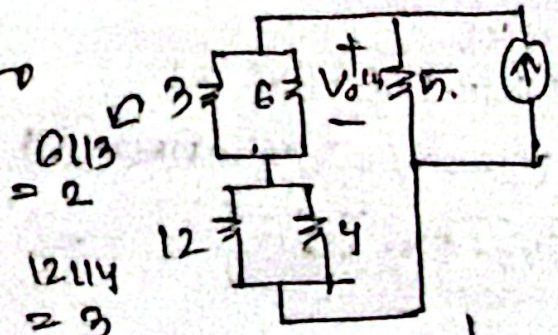
$V_3 = 2.85V$

$\therefore V_0' = -2.85$

$= -7.125V$.

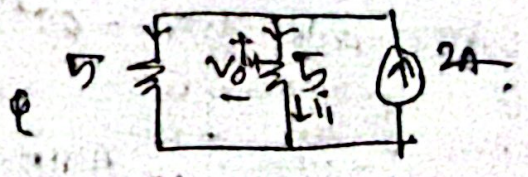
When only 2A is active,





$6 \| 3 = 2$

$12 \| 4 = 3$



$i_1 = \frac{5}{5+5} \times 2 = 1A$.

$V_0'' = i_1 \times 5 = 5V$.

$\therefore V_0 = V_0' + V_0'' + V_0'''$

$= 2 - 7.125 + 5$

$= -0.125V$

$= -125mV$.